

Generation Interconnection Request System Impact Study Report

For Queue Project Q22-2040

John S Rainey 100 kV
May 21, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2040. The Interconnection Customer proposes a 55.2 MW solar photovoltaic facility interconnecting at the John S Rainey 100 kV substation in the Indiana area. The request was received on March 22, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 435 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 14 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$26,690,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: POINT A - LAKE MARION 100kV
Contingency: RICHLAND - DEKALB 500kV
Loading: 158.7% (Rating: 224 MVA)
DFAX: 6.7%
Pre-project loading: 157.1%
2. Facility: POINT A SUBSTATION - BARRET 100kV L
Contingency: DEKALB - RICHLAND 500kV L
Loading: 153.0% (Rating: 247 MVA)
DFAX: 6.7%
Pre-project loading: 151.5%
3. Facility: SEGS 5 SUBSTATION - PEAVINE 100kV
Contingency: SEGS 5 - STATE ST 100kV Line
Loading: 150.3% (Rating: 203 MVA)
DFAX: 10.5%
Pre-project loading: 147.5%
4. Facility: GRABILL - JOHN S RAINEY 100kV
Contingency: ROSEAU COUNTY - S WHITEHALL 100kV Line
Loading: 142.4% (Rating: 127 MVA)
DFAX: 18.3%
Pre-project loading: 134.5%
5. Facility: ROSEAU COUNTY - BARRET 100kV
Contingency: DEKALB - RICHLAND 500kV L
Loading: 135.2% (Rating: 234 MVA)
DFAX: 12.3%
Pre-project loading: 132.3%

6. Facility: ORINDA - GRABILL 100kV
Contingency: ROSEAU COUNTY - S WHITEHALL 100kV Line
Loading: 134.8% (Rating: 118 MVA)
DFAX: 18.3%
Pre-project loading: 126.2%
7. Facility: ROSEAU COUNTY - ORINDA 100kV
Contingency: ROSEAU COUNTY - SOUTH WHITEHALL 100kV
Loading: 128.9% (Rating: 111 MVA)
DFAX: 18.3%
Pre-project loading: 119.7%
8. Facility: MONROEVILLE - WOODBURN 100kV
Contingency: RICHLAND - DEKALB 500kV
Loading: 125.0% (Rating: 277 MVA)
DFAX: 6.6%
Pre-project loading: 123.7%
9. Facility: SEGS 5 - STATE STREET 100kV
Contingency: RICHLAND - DEKALB 500kV
Loading: 124.8% (Rating: 386 MVA)
DFAX: 6.8%
Pre-project loading: 123.8%
10. Facility: WOODBURN - SAGE 100kV
Contingency: DEKALB - RICHLAND 500kV Line
Loading: 123.5% (Rating: 291 MVA)
DFAX: 6.6%
Pre-project loading: 122.2%
11. Facility: MONTE DEL ESTADO 6303 - SAGE 100kV
Contingency: RICHLAND - DEKALB 500kV Line
Loading: 121.5% (Rating: 311 MVA)
DFAX: 6.6%
Pre-project loading: 120.4%
12. Facility: HILLCREST - MULDOON MILL 100kV
Contingency: WAYNE TRACE - DARLINGTON RD 100kV line
Loading: 113.8% (Rating: 435 MVA)
DFAX: 7.9%
Pre-project loading: 112.8%
13. Facility: HARPER - MONTE DEL ESTADO 6303 100kV
Contingency: BARRET - ROSEAU COUNTY 100kV
Loading: 109.3% (Rating: 423 MVA)
DFAX: 7.5%
Pre-project loading: 108.3%
14. Facility: WAYNE TRACE - ALLEN 100kV
Contingency: SEGS 5 - HILLCREST 100kV line
Loading: 108.7% (Rating: 390 MVA)
DFAX: 6.6%
Pre-project loading: 107.8%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
POINT A - LAKE MARION 100kV	Rebuild line with higher-capacity conductor	\$4,738,000	38% (\$1,802,000)
POINT A SUBSTATION - BARRET 100kV L	Rebuild line with higher-capacity conductor	\$10,038,000	38% (\$3,819,000)
SEGS 5 SUBSTATION - PEAVINE 100kV	Rebuild line with higher-capacity conductor	\$1,523,000	2% (\$32,000)
GRABILL - JOHN S RAINEY 100kV	Rebuild line with higher-capacity conductor	\$2,641,000	100% (\$2,641,000)
BARRET - ROSEAU COUNTY 100kV	Rebuild line with higher-capacity conductor	\$9,739,000	38% (\$3,730,000)
ORINDA - GRABILL 100kV	Rebuild line with higher-capacity conductor	\$9,244,000	100% (\$9,244,000)
ORINDA SUBSTATION - ROSEAU COUNTY 100kV	Rebuild line with higher-capacity conductor	\$2,870,000	100% (\$2,870,000)
MONROEVILLE - WOODBURN 100kV	Rebuild line with higher-capacity conductor	\$4,784,000	20% (\$955,000)
SEGS 5 - STATE STREET 100kV	Rebuild line with higher-capacity conductor	\$2,830,000	1% (\$39,000)
WOODBURN - SAGE 100kV	Rebuild line with higher-capacity conductor	\$3,228,000	20% (\$645,000)
MONTE DEL ESTADO 6303 - SAGE 100kV	Rebuild line with higher-capacity conductor	\$3,785,000	20% (\$756,000)
HILLCREST - MULDOON MILL 100kV	Rebuild line with higher-capacity conductor	\$5,544,000	2% (\$88,000)
HARPER - MONTE DEL ESTADO 6303 100kV	Reconductor line	\$1,642,000	2% (\$38,000)
WAYNE TRACE - ALLEN 100kV	Reconductor line	\$2,273,000	1% (\$31,000)
	Total allocated to project		\$26,690,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.